Using minmax principle, determine the best alternative.

Solution:

Alternative	Pay-offs (Conditional events)				Maximum pay off
	A_1	A_2	A_3	A_4	
E_1	7	12	20	27	27
E_2	10	9	10	25	25
E_3	23	20	14	23	23
E_4	32	24	21	17	32

min(27, 25, 23, 32) = 23. Since the minimum cost is 23, the best alternative is E_3 according to minimax principle.

Exercise 10.3

1. Given the following pay-off matrix(in rupees) for three strategies and two states of nature.

Strategy	States-of-nature	
	E_1	E_2
S_1	40	60
S_2	10	–20
S_3	–40	150

Select a strategy using each of the following rule (i) Maximin (ii) Minimax.

2. A farmer wants to decide which of the three crops he should plant on his 100-acre farm. The profit from each is dependent on the rainfall during the growing season. The farmer has categorized the amount of rainfall as high medium and low. His estimated profit for each is shown in the table.

Rainfall	Estimated Conditional Profit(Rs.)		
	Crop *A*	Crop *B*	Crop *C*
High	8000	3500	5000
Medium	4500	4500	5000
Low	2000	5000	4000

If the farmer wishes to plant only crop, decide which should be his best crop using (i) Maximin (ii)Minimax

3. The research department of Hindustan Ltd. has recommended to pay marketing department to launch a shampoo of three different types. The marketing types of shampoo to be launched under the following estimated pay-offs for various level of sales.

Types of shampoo	Estimated Sales (in Units)		
	15000	10000	5000
Egg shampoo	30	10	10
Clinic Shampoo	40	15	5
Deluxe Shampoo	55	20	3

What will be the marketing manager's decision if (i) Maximin and (ii) Minimax principle applied?

4. Following pay-off matrix, which is the optimal decision under each of the following rule (i) maxmin (ii) minimax

Act	States of nature			
	S_1	S_2	S_3	S_4
A_1	14	9	10	5
A_2	11	10	8	7
A_3	9	10	10	11
A_4	8	10	11	13

Exercise 10.4

Choose the correct Answer

1. The transportation problem is said to be unbalanced if _____.
   (a) Total supply ≠ Total demand
   (b) Total supply = Total demand
   (c) $m = n$
   (d) $m+n-1$

2. In a non – degenerate solution number of allocations is
   (a) Equal to $m+n-1$
   (b) Equal to $m+n+1$

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